

Giuseppe Bassi

Last updated: August 2026

✉ giuseppe.bassi@princeton.edu

🌐 giuseppebassi.github.io

🐙 github.com/GiuseppeBassi

🌐 linkedin.com/in/giuseppe-bassi

Education

Princeton University

Ph.D. in Operations Research and Financial Engineering

Sep 2026 -

Princeton, NJ, USA

Bocconi University

B.Sc. in Mathematical and Computing Sciences for Artificial Intelligence

Sep 2023 - Jul 2026

Milan, Italy

- Graduated **110/110 cum laude**.
- Bachelor thesis: *Optimal Transport at Finite Temperature: A Rigorous Analysis of a Statistical-Physics Derivation*, supervised by Prof. Giuseppe Savaré.
- **Selected coursework:** probability; mathematical statistics; complex analysis and differential geometry; real and Fourier analysis; advanced optimization; mathematical modelling for machine learning; deep learning; statistical and quantum physics.

University of Toronto

Exchange Student

Sep 2025 - Dec 2025

Toronto, Canada

- Completed coursework in *Topology*, *Partial Differential Equations*, and *Statistical Computation*, earning A+ in all three courses.

Università degli Studi di Milano

Non-degree coursework

2025

Milan, Italy

- Attended courses in *Measure Theory*, *Functional Analysis*, and *Numerical Analysis*.

Liceo Scientifico Nicolò Copernico

Scientific High School Diploma

Sep 2018 - Jun 2023

Brescia, Italy

- Final grade: **100/100 cum laude**.

Academic Interests

High-dimensional probability and statistics; asymptotic analysis; convex optimization and duality; mathematical statistical mechanics; mathematical foundations of machine learning, with particular interest in concentration phenomena, random matrices, free-energy methods, and minimax theory.

Research Experience

Diffusion Models for Formal Mathematical Reasoning

Supervised by Dr. Massimiliano Esposito, Axiom Math; formerly IBM Research

Jul 2025 - Present

- Lead research on **masked diffusion for automated Lean proof repair**, investigating how iterative denoising can be used to repair invalid formal proofs and whether it provides advantages over autoregressive approaches.
- Study the mechanisms governing successful proof repair, including the role of Lean verifier feedback, proof states, corruption geometry, and decoding dynamics. Adapting **Pythagoras-Prover-Diffusion-4B** to proof repair on the **APRIL** dataset under controlled comparisons.
- Further developments: integrate the Corrector with a diffusion-based proof generator and Lean verifier into an iterative **Prover, Verifier, Corrector** architecture, and train the system with reinforcement learning using formal verification as the reward signal.

Bachelor Thesis: Optimal Transport at Finite Temperature

Advisor: Prof. Giuseppe Savaré, Bocconi University

Mar 2025 - Jul 2026

Milan, Italy

- Formalized the finite-temperature optimal transport construction of Koehl et al. in a **convex-analytic framework**, providing original proofs of its main analytical properties independently of the heuristic statistical-physics derivation.
- Developed an original **duality theory** for the finite-temperature algorithm: derived a primal regularized transport problem whose optimality equations recover the finite-temperature system, and showed that its Lagrangian dual is the effective free energy, revealing a variational parallel with Cuturi's entropic optimal transport.
- Reconstructed the statistical-physics derivation using **geometric measure theory**, Fourier and complex analysis:

defined integration intrinsically on the transport polytope, replaced formal Dirac constraints using the coarea formula and Gaussian approximations, and isolated the analytic hypotheses required for a temperature-dependent multidimensional **steepest-descent analysis**.

- Reassessed the mathematical claims of the original physics treatment, proving some, identifying missing hypotheses in others, and disproving a claimed triangle inequality for the finite-temperature free energy. Ongoing work investigates further metric claims and sharper probabilistic bounds and asymptotics in the finite-temperature regime.

London Geometry and Machine Learning (LOGML)

Jul 2025

Supervised by Dr. Michael Murray, University of Bath

London, UK

- Investigated storage capacity and emergent invariance in **Hopfield networks** trained by energy-flow minimization on graph-orbit datasets, studying when a network trained on a small sample of graph isomorphisms can memorize the full isomorphism class.
- Recast Hopfield stability on graph orbits as linear separability constraints, relating energy-flow minimization to a **hard-margin SVM** and exploiting graph-orbit symmetry.
- Formulated the resulting sample-complexity problem as controlling the deviation between the SVM trained on a random subset of the orbit and the full-orbit solution, and investigated concentration-based approaches to probabilistic generalization bounds.

Teaching Experience & Mathematical Outreach

Teaching Assistant, Mathematical Analysis II

Feb 2025 - Jun 2025

Bocconi University

Milan, Italy

- Led undergraduate exercise sessions under the supervision of Prof. Federico Vegni. Prepared problem sets and exercise materials.

Mathematical Olympiad Instructor

Nov 2023 - Apr 2025

Liceo Scientifico "Niccolò Copernico" - PNRR STEM Olympiad Mathematics Program

Brescia, Italy

- Designed and taught 20 hours of training across two editions of the program for selected students from the Province of Brescia, preparing them for the national finals of the Italian Mathematical Olympiad team competition.
- Taught at two intensive Olympiad training camps in Capo di Ponte.

Educational Materials

2025 - Present

giuseppegbassi.github.io

- Maintain a public collection of mathematical notes and Olympiad materials; the main project is a 185-page set of *Probability Theory* notes developed from measure theory and Fourier analysis.

Honors & Scholarships

Bocconi Excellence Award

2023-24, 2024-25

Bocconi University

- Full-tuition merit scholarship of approximately **€17,000 per year**, awarded by GPA to the **top 15 of about 2,500 students**. Received in both years of eligibility.

Fondazione Paolo e Carolina Zani Scholarship

2023-2026

Province of Brescia

- Three-year merit scholarship of **€5,000 per year**, awarded to two students across the Province of Brescia for exceptional secondary-school academic achievement; about 10,000 students graduate in the province each year.

Nobilini-Bahlke Scholarship

2023

Liceo Scientifico "Niccolò Copernico", Brescia

- Merit award for the two graduating students with the strongest academic performance over five years.

Mathematical Competitions

William Lowell Putnam Mathematical Competition **2025**
University of Toronto

- Ranked **179th out of 4,329 participants**; score 43/120.

Imperial-Cambridge Mathematics Competition **2025**

- Honorable mention, ranked **39th** overall.

Italian Mathematical Olympiad **2018 - 2023**

- Two **silver medals** in the individual national finals (2022, 2023).
- **Bronze medal** as team captain in the 2023 national team competition.
- Selected with full financial support for two national training and selection camps at Scuola Normale Superiore in Pisa: **Stage Senior** (~80 students) and **Winter Camp** (~30 students), part of Italy's IMO selection pathway.

International Mathematical Competitions **2022 - 2023**

- **Iranian Combinatorics Olympiad (ICO)**: selected to represent Italy; **silver medal**.
- **Iranian Geometry Olympiad (IGO)**: selected to represent Italy.

Other National Mathematical Competitions **2017 - 2023**

- **Gran Premio della Matematica Applicata**: 2nd place out of more than 10,000 participants (2023).
- **Kangourou della Matematica**: three-time national finalist, placing 22nd, 6th, and 11th.

Academic Leadership & Service

Founder and President, Bocconi Student Association for Mathematical Sciences (BAMS) **Sep 2025 - Jul 2026**

Bocconi University

Milan, Italy

- Founded Bocconi's first student association devoted to the mathematical sciences, **creating an academic community** for students interested in mathematics, statistics, applied mathematics and physics, and organizing problem-solving sessions, reading groups, seminars, and long-term student projects.
- Worked with Prof. Alessandro Pigati to introduce the **Imperial-Cambridge Mathematics Competition (ICMC)** at Bocconi and led problem-solving preparation sessions.

Member, Bocconi AI and Neuroscience Student Association (BAINSA) **Feb 2024 - Jul 2026**

AI Research Team, Bocconi University

Milan, Italy

- Contributed to the *Cryptocurrency Forecasting* project, studying the impact of news and social-media sentiment on price dynamics using fine-tuned LLMs and LSTM-based models; developed the text-processing and sentiment-extraction pipeline and fine-tuned an LLM for news classification.

Skills

Languages

- Italian: native.
- English: C1 (IELTS, Jan 2025).

Programming languages

- Python (PyTorch, NumPy, pandas, scikit-learn) | R | C

Formal methods

- Lean

Markup and research tools

- L^AT_EX | Git